

# Mohammad Nour

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## PROFESSIONAL SUMMARY

Third-year PhD researcher specializing in hardware security and machine learning for cryptographic analysis. Experienced in side-channel attacks, deep learning for key recovery, and explainable AI. Strong background in cryptography and ML, with hands-on expertise in PyTorch, TensorFlow, and scikit-learn.

## SKILLS

**Programming Languages:** Verilog, Python, C/C++, MATLAB, Shell scripting, solidity.

**DL/ML:** PyTorch, PyTorch Lightning; TensorFlow (v1/v2); OpenCV; scikit-learn; NumPy, Pandas.

**Platforms & Tooling:** Debian/GNU Linux, Windows; Git, Docker, bash; basic CUDA.

## PROJECTS

### There's Waldo: PCB Tamper Forensic Analysis using Explainable AI on Impedance Signatures

*IEEE International Symposium on Electromagnetic Compatibility, Signal and Power Integrity (EMC+SIPI), 2025*

- Developed an explainable machine-learning framework that uses impedance (S-parameter) signatures to detect and classify PCB tampering with 96% accuracy.

### Uncertainty Estimation in Neural Network-enabled Side-channel Analysis

*Conference on Cryptographic Hardware and Embedded Systems (CHES), 2026*

- Developed a framework that quantifies predictive uncertainty in neural-network side-channel attacks, separating contributions from model behavior and data quality.

### Too hot to be true: Temperature calibration for NN-assisted side-channel analysis

*Constructive Approaches for SeCurity Analysis and Design of Embedded systems (CASCADE), 2026*

- Studied the effects of temperature calibration on deep learning based SCA.

### Swarm in EM Hay: Particle Swarm-guided Probe Placement for EM SCA

*USENIX WOOT Conference on Offensive Technologies, 2026*

- Studied the effects of temperature calibration on deep learning based SCA.

### Rock and a Hard Place: Attack Hardness in Neural Network-assisted Side Channel Analysis

*under review*

## SOCIAL ACTIVITIES AND LEADERSHIP

- Leading the Iranian Student Association at Worcester Polytechnic Institute (2023–2025).
- Organizing and founding the OPTIMIST PhD forum on AI and side-channel analysis.
- Mentoring four undergraduate researchers** and promoted reproducibility by releasing six open-source GitHub repositories supporting research artifacts.

## AWARDS AND HONORS

- Recipient of the **DAC Young Fellow Award 2025**.
- Invited talk at **New England Hardware Security Day 2025**, MIT.
- 3rd Place Best Poster Award** at New England Hardware Security Day 2024, WPI.

## EDUCATION

### Worcester Polytechnic Institute

2023 - Present

*PhD - Electrical and Computer Engineering | Advisor: Prof. Fatemeh Ganji*

Worcester, MA

*Courses: Applied cryptography & physical attacks, Physical Security of Microelectronic Systems*

### Sharif University of Technology

2020 - 2023

*M.Sc. - secure communications and cryptography*

*Courses: Deep learning, Cryptography, foundations of the blockchains*